Ravirala Aditya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.56
Narayana junior college	Class XII (Telangana State Board of Intermediate education(TSBIE))	2023	97.80%
Narayana high school	Class X (Secondary School Certificate Board(SSC))	2021	10.00%

WORK EXPIRENCE

Software Engineering Intern -- *Segment (Twilio)*

- **AI-Powered Development Pipeline:** Built a multi-agent automation workflow using **Claude Code Skills** to generate production-ready cloud integrations directly from API documentation, cutting average turnaround time from 6-7 weeks down to 1-2 weeks (**70-80% reduction**).
- **Two-Layer E2E Testing:** Created local and staging end-to-end test generators that execute directly inside remote cluster pods, verifying real-time payload delivery against live AWS services.
- **Production Runtime Environment:** Set up containerized Kubernetes pods with least-privilege access to support autonomous code generation, automated branch creation, and pull request workflows.
- **Quality & Benchmarking:** Authored manual reference integrations to run gap-analysis against AI-generated code, iterated over agent rules until the pipeline consistently produced zero-gap codebases.

PROJECTS

Ball Trees Data Structure & Nearest Neighbor Analysis (*Github link*

- **Core Data Structure:** Implemented high-dimensional Ball Trees in C++ and Python supporting exact, approximate k-NN, and spatial range queries across N-dimensional spaces.
- **Algorithmic Optimization:** Integrated **Welzl's Algorithm** for minimum enclosing ball calculations, significantly optimizing tree bounding volumes, node partitioning efficiency, and spatial query pruning.
- **Performance Benchmarking & Analysis:** Constructed a custom C++ testbench comparing Ball Trees against naive k-NN baseline searches across varied tree depths, tracking points searched vs. skipped to demonstrate clear logarithmic speedups.
- **Visualization & Applications:** Developed 2D spatial partitioning visualizations using Matplotlib and applied the data structure to high-dimensional image classification workflows.

Real-Time Chat application (*Github link*)

- Built a real-time chat application using React, Node.js, Express.js, and MongoDB; added Socket.IO for messaging.
- Designed the authentication flow with Express.js and MongoDB, storing sessions client-side via localStorage and hashing credentials using bcrypt. Verified users through RESTful API endpoints for login and registration.
- Designed responsive interfaces with React, featuring emoji support and message synchronization with Socket.IO

Agentic Linux Dlagnostic Assistant (*Github link*)

- Built an LLM agent that diagnoses Linux/service/Git issues via 18 tools exposed through a custom MCP server, orchestrated by Gemini & LangGraph
- Designed a risk-tiered permission system with human-in-the -loop approval to saftey constrain LLM driven system actions.

Statistical Data Analysis (*Github link*)

- Performed EDA on multiple real world datasets using statistical methods to derive correlations, spot trends, and

visualize insights through bar charts, histograms, box plots, pie charts.
- Applied Central Limit Theorem on Domino's sales data to estimate the avg daily orders and sales, using sampling techniques to analyze the behaviour of sample means.
- Modeled IPL players strike rates using gamma distribution and estimated the gamma parameters using MLE and MOM estimators and performed hypothesis testing on IPL, built confidence intervals.

SKILLS

Languages: Proficient in C, C++, Familiar With python, JavaScript, HTML, CSS, Node.js, Express.js, React.js, Socket.IO

Cloud & DevOps: AWS (Bedrock, Kinesis, S3, STS, IAM), Kubernetes (K8s), Docker

Tools: GitHub, Claude Code CLI (Skills), Claude Code Skills, Jest,, LaTeX, SQL

RELEVANT COURSES

Mathematics: Calculus, Introduction to Number theory, Combinatorics and Graph theory, Convex optimization, Ordinary Differential Equations, Linear Algebra, Probability, Applied Statistics, Transform Techniques, Groups and Rings, Writing and Presenting Mathematics, Real analysis, Numerical analysis

Computer Science: Introduction to Programming, Data Structures, Algorithms, DBMS-1, Compilers-1, Theory of computation

EXTRACURRICULAR

- National Service Scheme (**NSS**) -Student Volunteer